CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ SOLUTIONS

Chapter 9: Some Applications of Trigonometry

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 One right-angled triangle

1. A car drives away from the foot of a tower 30 m high. What is the angle of elevation of the top of the tower from the car when it is $10\sqrt{3}$ m from the foot? (a) 30° (b) 45° (c) 90° (d) 60° CBSE 2026, 1 mark

SOLUTION

$$\tan\theta = \frac{\text{height}}{\text{distance}} = \frac{30}{10\sqrt{3}} = \frac{3}{\sqrt{3}} = \sqrt{3}$$

$$\tan 60^\circ = \sqrt{3}$$

Answer: (d)

2. A kite is flying 150 m above the ground. Its string is straight and makes an angle of 30° with the ground. How long is the string? CBSE 2025, 1 mark

SOLUTION

The height is the side opposite the 30° angle and the string is the hypotenuse.

$$\sin 30^\circ = \frac{150}{\text{string}}, \text{ so string} = \frac{150}{\frac{1}{2}}$$

Answer: 300 m

3. A ladder leaning against a wall makes an angle of 60° with the ground. Its foot is 2.5 m from the wall. Find the length of the ladder. CBSE 2016, 1 mark

SOLUTION

The distance from the wall is the side adjacent to the 60° angle, and the ladder is the hypotenuse.

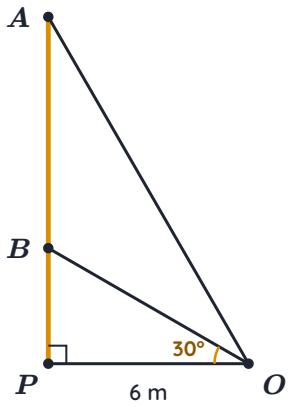
$$\cos 60^\circ = \frac{2.5}{\text{ladder}}, \text{ so ladder} = \frac{2.5}{\frac{1}{2}}$$

Answer: 5 m

MathSetu

2 Two right-angled triangles

4. A radio tower PA is built in two sections: PB is the lower section and BA the upper one. Wires from a point O on the ground, 6 m from the foot P, hold the tower at B and at A. From O the angle of elevation of B is 30° and that of A is 60°. (i) Find the length of the wire OB. (ii) Find the length of the wire OA. (iii) Find AB, OR find the area of $\triangle OPB$. CBSE 2026, 4 marks



Angle POB = 30° and angle POA = 60°

SOLUTION

(i) In $\triangle OPB$, $\cos 30^\circ = \dfrac{OP}{OB}$, so $OB = \dfrac{6}{\frac{\sqrt{3}}{2}} = \dfrac{12}{\sqrt{3}}$. $OB = 4\sqrt{3}$ **m.**

(ii) In $\triangle OPA$, $\cos 60^\circ = \dfrac{OP}{OA}$, so $OA = \dfrac{6}{\frac{1}{2}}$. $OA = 12$ **m.**

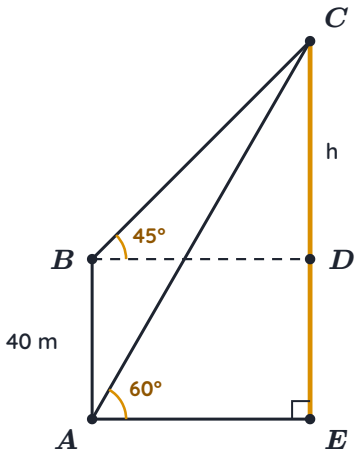
(iii) $PB = 6\tan 30^\circ = \dfrac{6}{\sqrt{3}} = 2\sqrt{3}$ m and $PA = 6\tan 60^\circ = 6\sqrt{3}$ m.

$AB = PA - PB = 6\sqrt{3} - 2\sqrt{3}$. $AB = 4\sqrt{3}$ **m.**

OR: area of $\triangle OPB = \dfrac{1}{2} \times OP \times PB = \dfrac{1}{2} \times 6 \times 2\sqrt{3}$. **Area** $= 6\sqrt{3}$ **m².**

MathSetu

5. From a point A on the ground, the top C of a lighthouse CE is seen at an angle of elevation of $60°$. From a deck at B, 40 m directly above A, the angle of elevation of C is $45°$. D is the point on CE level with B, and $CD = h$ metres. (i) Write BD in terms of h. (ii) Write BC in terms of h. (iii) Find the height CE, taking $\sqrt{3} = 1.73$, OR find AE if $AC = 100$ m. CBSE 2025, 4 marks



SOLUTION

(i) In $\triangle BDC$, $\tan 45° = \dfrac{CD}{BD}$, so $BD = CD$. $BD = h$ **metres.**

(ii) $\sin 45° = \dfrac{CD}{BC}$, so $BC = \dfrac{h}{\frac{1}{\sqrt{2}}}$. $BC = \sqrt{2}\,h$ **metres.**

(iii) $AE = BD = h$ and $CE = h + 40$. In $\triangle AEC$, $\tan 60° = \dfrac{h + 40}{h}$.

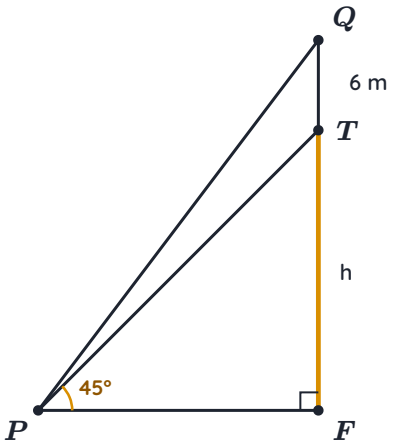
$\sqrt{3}\,h = h + 40$, so $h = \dfrac{40}{\sqrt{3} - 1} = \dfrac{40(\sqrt{3} + 1)}{2} = 20(\sqrt{3} + 1) = 20 \times 2.73 = 54.6$

$CE = 54.6 + 40$. **The lighthouse is 94.6 m high.**

OR: in $\triangle AEC$, $\cos 60° = \dfrac{AE}{AC}$, so $AE = 100 \times \dfrac{1}{2}$. $AE = 50$ **m.**

MathSetu

6. A 6 m pole stands on top of a tower. From a point P on the ground, the top of the pole is seen at an angle of elevation of 60°. From the top of the tower, P is seen at an angle of depression of 45°. Find the height of the tower and the distance of P from its foot. (Take $\sqrt{3} = 1.73$.) CBSE 2024, 5 marks



FT is the tower and TQ is the pole

SOLUTION

Let the height of the tower be h m and the distance of P from its foot be d m.

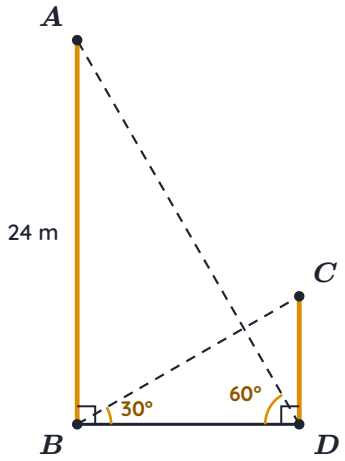
The angle of depression of P from the top of the tower is 45°, so the angle of elevation of the top of the tower from P is 45°: $\tan 45^\circ = \dfrac{h}{d}$, so $d = h$.

For the top of the pole: $\tan 60^\circ = \dfrac{h+6}{d}$, so $\sqrt{3}\,h = h + 6$.

$$h = \frac{6}{\sqrt{3}-1} = \frac{6(\sqrt{3}+1)}{2} = 3(\sqrt{3}+1) = 3 \times 2.73 = 8.19$$

Answer: the tower is 8.19 m high and P is 8.19 m from its foot.

7. Two towers stand on level ground. The first is 24 m high, and from the foot of the second tower its top is seen at an angle of elevation of 60°. From the foot of the first tower, the top of the second is seen at an angle of elevation of 30°. Find the distance between the towers, the height of the second tower, and the length of a wire joining the two tops. CBSE 2023, 5 marks



SOLUTION

Let $AB = 24$ m be the first tower, $CD = h$ the second, and $BD = d$ the distance between them.

In $\triangle ABD$: $\tan 60^\circ = \dfrac{24}{d}$, so $d = \dfrac{24}{\sqrt{3}} = 8\sqrt{3}$ m ≈ 13.86 m.

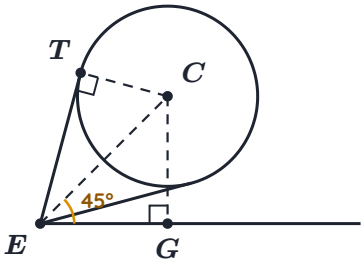
In $\triangle CDB$: $\tan 30^\circ = \dfrac{h}{d}$, so $h = \dfrac{8\sqrt{3}}{\sqrt{3}} = 8$ m.

The tops differ in height by $24 - 8 = 16$ m. Wire $AC = \sqrt{(8\sqrt{3})^2 + 16^2} = \sqrt{192 + 256} = \sqrt{448} = 8\sqrt{7}$ m.

Answer: distance $8\sqrt{3}$ m; second tower 8 m; wire $8\sqrt{7}$ m ≈ 21.17 m

Mathsetu

8. A spherical balloon of radius r subtends an angle of 60° at the eye of an observer, and the angle of elevation of its centre from the eye is 45°. Prove that the centre of the balloon is at a height of $\sqrt{2}\,r$.
CBSE 2023, 5 marks



SOLUTION

Let E be the eye, C the centre of the balloon, ET a tangent to the balloon and CG the height of the centre above the level of the eye.

The two tangents from E make an angle of 60°, and EC bisects it, so $\angle CET = 30^\circ$.

$CT \perp ET$ (radius and tangent). In $\triangle CTE$: $\sin 30^\circ = \dfrac{r}{EC}$, so $EC = 2r$.

In $\triangle CGE$: $\sin 45^\circ = \dfrac{CG}{EC}$, so $CG = 2r \times \dfrac{1}{\sqrt{2}} = \sqrt{2}\,r$.

So the height of the centre is $\sqrt{2}$ times the radius. Proved.

9. A tower is 50 m high. From the foot of a building, the top of the tower is seen at an angle of elevation of 60°. From the foot of the tower, the top of the building is seen at an angle of elevation of 30°. Find the height of the building. CBSE 2022 Term 2, 3 marks

SOLUTION

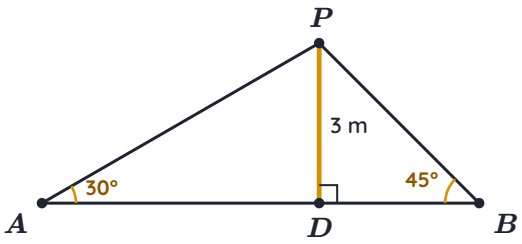
Let the distance between them be d and the height of the building be h.

$\tan 60^\circ = \dfrac{50}{d}$, so $d = \dfrac{50}{\sqrt{3}}$ m.

$\tan 30^\circ = \dfrac{h}{d}$, so $h = \dfrac{50}{\sqrt{3}} \times \dfrac{1}{\sqrt{3}} = \dfrac{50}{3}$

Answer: $16\dfrac{2}{3}$ m, about 16.67 m

10. From a point on a bridge 3 m above the banks of a river, the two banks on opposite sides are seen at angles of depression of 30° and 45°. Find the width of the river. CBSE 2022 Term 2, 3 marks



SOLUTION

Let P be the point on the bridge, D the point directly below it, and A and B the two banks. The angles of depression equal the angles at A and B (alternate angles).

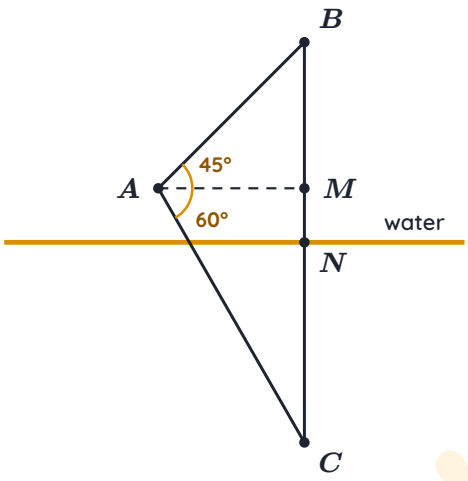
$\tan 30^\circ = \dfrac{3}{AD}$, so $AD = 3\sqrt{3}$ m. $\tan 45^\circ = \dfrac{3}{DB}$, so $DB = 3$ m.

Width $= AD + DB = 3\sqrt{3} + 3$

Answer: $3(\sqrt{3}+1)$ **m, about 8.2 m**

MathSetu

11. The top B of a pavilion standing in a lake is 10 m above the water. From a point A, h metres above the water, the angle of elevation of B is 45° and the angle of depression of the reflection C of B in the water is 60°. (i) Draw a labelled figure. (ii) Find h, taking $\sqrt{3} = 1.73$. CBSE 2022 Term 2, 4 marks



N is on the water surface, with BN = NC = 10 m

SOLUTION

(i) The reflection C is as far below the water surface as B is above it, so $NC = BN = 10$ m. Draw AM horizontal, meeting BC at M. Then $MN = h$.

(ii) $BM = 10 - h$ and $MC = 10 + h$.

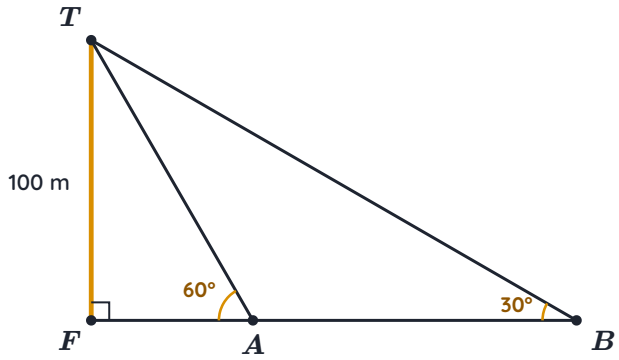
In $\triangle AMB$: $\tan 45^\circ = \dfrac{10-h}{AM}$, so $AM = 10 - h$.

In $\triangle AMC$: $\tan 60^\circ = \dfrac{10+h}{AM}$, so $\sqrt{3}(10-h) = 10 + h$.

$h(\sqrt{3}+1) = 10(\sqrt{3}-1)$, so $h = \dfrac{10(\sqrt{3}-1)}{\sqrt{3}+1} = \dfrac{10(\sqrt{3}-1)^2}{2} = 5(4-2\sqrt{3}) = 20 - 10\sqrt{3}$

Answer: $h = 20 - 17.3 = 2.7$ **m**

12. A man rows a boat directly away from a lighthouse 100 m high. In 2 minutes the angle of elevation of the top of the lighthouse changes from 60° to 30°. Find the speed of the boat in metres per minute. Take $\sqrt{3} = 1.732$. CBSE 2019, 4 marks



SOLUTION

Let TF be the lighthouse, and let the boat move from A to B.

$\tan 60^\circ = \dfrac{100}{FA}$, so $FA = \dfrac{100}{\sqrt{3}}$ m. $\tan 30^\circ = \dfrac{100}{FB}$, so $FB = 100\sqrt{3}$ m.

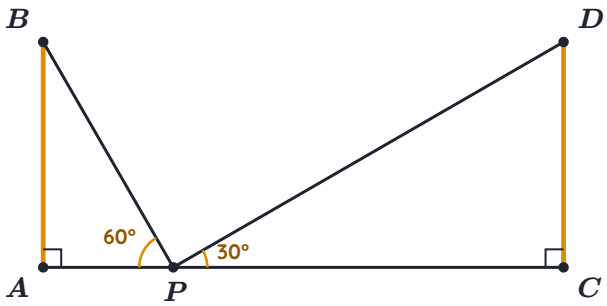
$$AB = 100\sqrt{3} - \frac{100}{\sqrt{3}} = \frac{300 - 100}{\sqrt{3}} = \frac{200}{\sqrt{3}} = \frac{200\sqrt{3}}{3} \approx 115.47 \text{ m}$$

$$\text{Speed} = \frac{115.47}{2}$$

Answer: about 57.73 m per minute

MathSetu

13. Two poles of equal height stand opposite each other on the two sides of a road 80 m wide. From a point on the road between them, the tops of the poles are seen at angles of elevation of 60° and 30°. Find the height of the poles and the distances of the point from them. CBSE 2019, 4 marks



SOLUTION

Let the poles AB and CD have height h, and let the point P be x m from the first pole, so it is $(80 - x)$ m from the second.

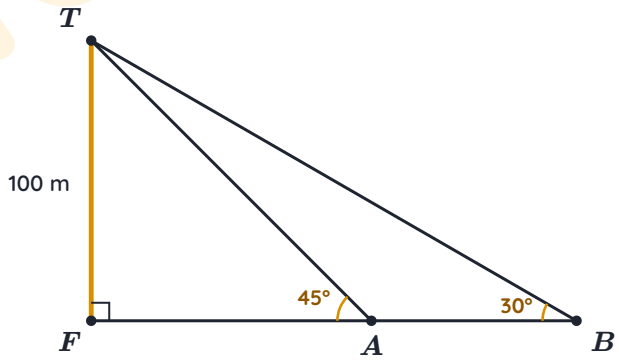
$\tan 60^\circ = \dfrac{h}{x}$, so $h = \sqrt{3}\,x$. $\tan 30^\circ = \dfrac{h}{80 - x}$, so $h = \dfrac{80 - x}{\sqrt{3}}$.

$\sqrt{3}\,x = \dfrac{80 - x}{\sqrt{3}}$, so $3x = 80 - x$ and $x = 20$.

$h = 20\sqrt{3} \approx 34.64$ m

Answer: each pole is $20\sqrt{3}$ m high; the point is 20 m from one pole and 60 m from the other

14. From the top of a lighthouse 100 m above sea level, two ships are seen at angles of depression of 30° and 45°. One ship is directly behind the other on the same side of the lighthouse. Find the distance between the ships. Take $\sqrt{3} = 1.732$. CBSE 2018, 4 marks



SOLUTION

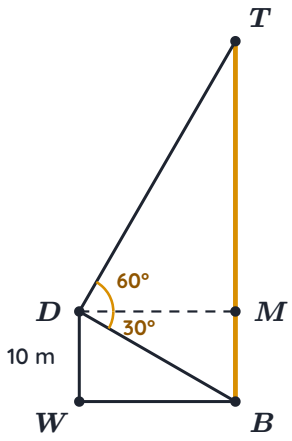
Let TF be the lighthouse and A, B the ships. The angles of depression equal the angles of elevation at the ships (alternate angles).

$\tan 45^\circ = \dfrac{100}{FA}$, so $FA = 100$ m. $\tan 30^\circ = \dfrac{100}{FB}$, so $FB = 100\sqrt{3}$ m.

$AB = 100\sqrt{3} - 100 = 100(1.732 - 1)$

Answer: 73.2 m

15. A man stands on the deck of a ship, 10 m above the water. He sees the top of a hill at an angle of elevation of 60° and the foot of the hill at an angle of depression of 30°. Find the distance of the hill from the ship and the height of the hill. CBSE 2016, 3 marks



SOLUTION

Let D be the man, W the point on the water below him, BT the hill and DM the horizontal through D. Then $MB = DW = 10$ m.

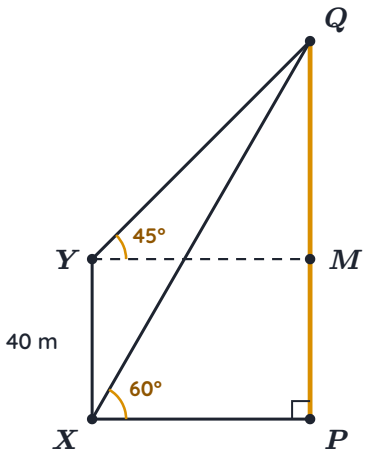
In $\triangle DMB$: $\tan 30^\circ = \dfrac{10}{DM}$, so $DM = 10\sqrt{3}$ m.

In $\triangle DMT$: $\tan 60^\circ = \dfrac{MT}{DM}$, so $MT = 10\sqrt{3} \times \sqrt{3} = 30$ m.

Height of the hill $= MT + MB = 30 + 10$

Answer: distance $10\sqrt{3}$ m (about 17.3 m); height 40 m

16. The top Q of a vertical tower PQ is seen at an angle of elevation of 60° from a point X on the ground. From a point Y, 40 m vertically above X, the angle of elevation of Q is 45°. Find the height PQ and the distance PX. Take $\sqrt{3} = 1.73$. CBSE 2016, 4 marks



SOLUTION

Draw YM horizontal, meeting PQ at M. Let $PX = YM = d$ and $PQ = h$, so $MQ = h - 40$.

In $\triangle YMQ$: $\tan 45^\circ = \dfrac{h-40}{d}$, so $d = h - 40$.

In $\triangle XPQ$: $\tan 60^\circ = \dfrac{h}{d}$, so $h = \sqrt{3}\,d$.

$d = \sqrt{3}\,d - 40$, so $d = \dfrac{40}{\sqrt{3}-1} = 20(\sqrt{3}+1) = 20 \times 2.73 = 54.6$

$h = 54.6 + 40 = 94.6$

Answer: $PQ = 94.6$ **m and** $PX = 54.6$ **m**

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